

Ensemble Techniques:

- * Combine diff ML Algorithms.

- * Two Techniques of Ensemble Learning
Bagging & Boosting:

I) Bagging: $\Rightarrow$ Random forest

- * In Bagging we have Multiple Base Learners (ML Algorithms), it can be same algorithm or diff.

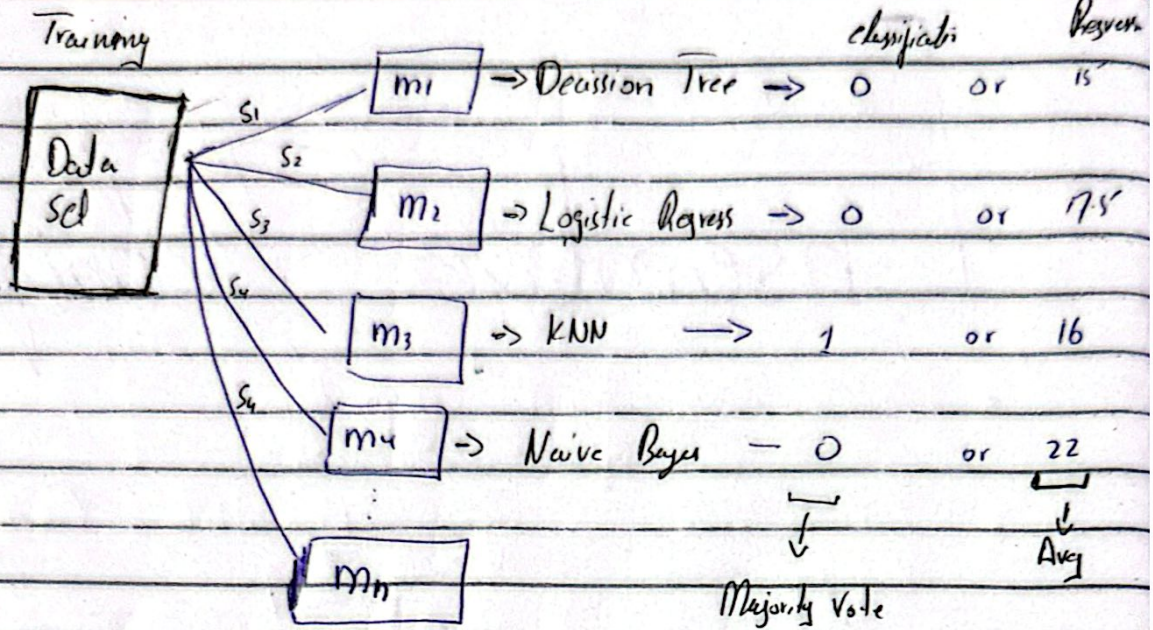
- * We give diff samples of Datasets to these models (learners)

- * Now Test data Arrives, give it to all the models, each will predict (for exm "3" predicted "Zero" & "5" models predicted "1") $\Rightarrow$ classification.

- * Go with majority Voting classifier.

- *** In Case of Regression final output will be average of all the models outputs.

output

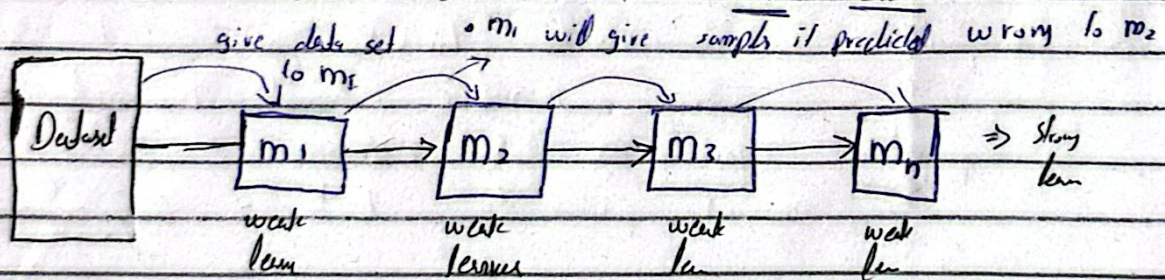


II) Boosting:

- Adaboost
- Gradient Boosting
- extreboost

* Models are connected sequentially.

* here Models are called weak learners



* Give entire dataset to model 1.

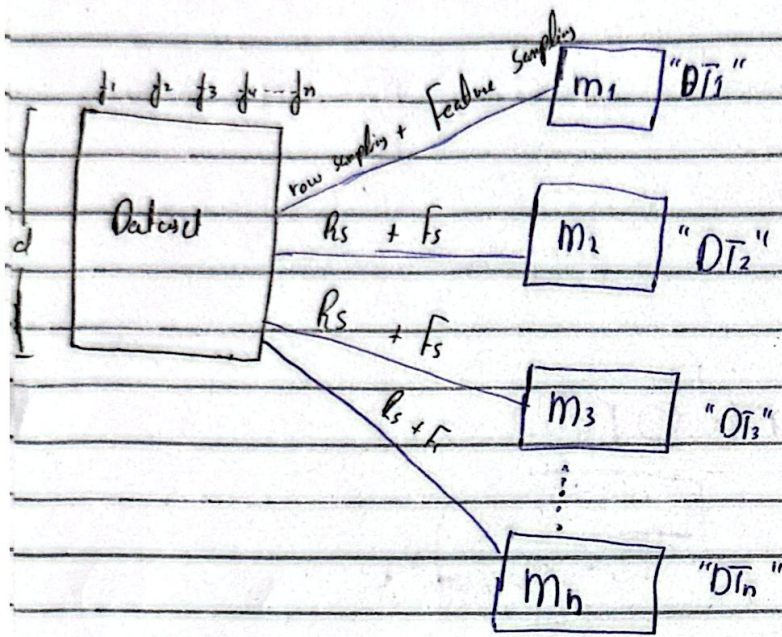
$\hookrightarrow$ some records that models didn't predict correctly will be given to model 2 & so on.

- ~ ~ ~ ~ ~
- * Combine multiple weak learners together to form a strong learner.

⇒ Random Forest:

- * Both classification & Regression
- * Use approach of Bagging technique
- * All the Base Models are Decision Trees.
- * We will give diff samples from datasets & give it to diff Base Models.
- ⇒ Row Sampling: Pick some of Rows
- ⇒ Feature Sampling: Pick some features.
- * So for every Base Model we will do Row sampling + Feature sampling.

R_s = row sampling
 F_s = Feature Sampling.



* Classification $\Rightarrow$ Majority Voting classifier

* Regression $\Rightarrow$ Avg output of the models.